

MD SOAD SOLAIMAN

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EDUCATION

Master of Science in Data Science

Jan 2025 – Present (Expected Nov 2026)

Adelaide University, Adelaide, Australia | GPA: 6.2 / 7.0

- Relevant coursework: Data Visualisation (High Distinction), Statistical Foundation on AIML, ML Algorithms, Deep Learning Fundamentals

Bachelor of Science in Industrial & Production Engineering

Jan 2020 – Mar 2024

Military Institute of Science and Technology (MIST), BUP, Bangladesh | CGPA: 3.69 / 4.00

RESEARCH & TECHNICAL PROJECTS

Finding Home in South Australia: Rental Affordability for International Students

2026

Project, Adelaide University

- Analysed six years (2020–2025) of SA Government Private Rent Report data across metropolitan zones, regional areas, dwelling types and bedroom categories.
 - Measured rental stress against the 30% and 50% income thresholds, showing that solo renting is unaffordable on a typical student income and that sharing is the only workable option in most regions.
 - Identified community-level pressures, including rapid post-COVID rent growth, a widening flat–house gap in regional SA and the resulting reliance on crowded, shared housing.
 - Built an interactive R/Quarto/Shiny affordability calculator for public use, with income, sharing and regional trend scenarios.
- [Live Report](#) | [Interactive Calculator](#) | [GitHub Repository](#)

Automating Document Outputs for New Hires V2

Aug 2026 – Present

Master's Capstone, Adelaide University with Career Embassy

- Investigate system requirements and map questionnaire structures and branching logic, building practical skills in survey design.
- Research Australian employment awards to support automated, compliant document generation.

Trustworthy Foundation Models for Time-Series Forecasting

2026

Research Project

- Developed a leakage-controlled, reproducible forecasting framework comparing Chronos, TimesFM, classical models (ARIMA, SARIMA, Holt-Winters, exponential smoothing), LSTM, Prophet and naive models across Bitcoin and South Australian electricity demand data.
 - Evaluated models on accuracy, statistical significance, robustness, temporal stability and uncertainty calibration.
- [GitHub Repository](#)

Statistical Research Analytics

2026

Projects

- Conducted a staged statistical workflow in R/Quarto on 2,392 records: exploratory analysis, ANOVA, Tukey HSD, t-tests, chi-squared tests, linear regression and residual diagnostics.
- [GitHub Repository](#)

MIT-BIH Arrhythmia Heartbeat Classification

2026

Coursework Research Project

- Built a record-level pipeline classifying four AAMI heartbeat classes, comparing Logistic Regression, Random Forest and CNN models.
 - Evaluated per-class precision, recall, F1 and PR-AUC, and used McNemar's test to compare minority-class recall.
- [Project Report](#) | [GitHub Repository](#)

Machine Learning & Deep Learning Portfolio

2026

Projects

- Built ResNet, CNN and MLP pipelines for image and tabular classification (blood-cell, butterfly and diabetes datasets), and applied PCA with K-Means for unsupervised clustering.
- [GitHub Repository](#)

TECHNICAL SKILLS

- Statistics and inference:** Hypothesis testing, ANOVA, Tukey HSD, t-tests, chi-squared tests, linear and logistic regression, bootstrap confidence intervals, model diagnostics
- Data preparation:** Cleaning messy public and administrative data, missing-value handling, reshaping, normalisation, feature selection and engineering, PCA, SVD, t-SNE

- **Programming and libraries:** R (tidyverse, tidymodels, caret); Python (pandas, NumPy, SciPy, scikit-learn, PyTorch, TensorFlow, Keras, OpenCV)
- **Machine learning:** Linear, ridge, lasso and logistic regression, decision trees, random forests, SVMs, K-means clustering, ensemble methods (bagging, boosting, AdaBoost, stacking)
- **Deep learning:** MLPs, CNNs, ResNets, RNNs, LSTMs, transfer learning, fine-tuning, self-supervised and contrastive learning
- **Forecasting:** ARIMA, SARIMA, exponential smoothing, Prophet, Chronos, TimesFM, uncertainty calibration
- **Model evaluation:** Cross-validation, precision, recall, F1, ROC-AUC, calibration, ablation studies, leakage prevention
- **Visualisation and reporting:** Tableau, Shiny, Plotly, Quarto, R Markdown, Microsoft Office
- **Research tools:** Git/GitHub, Jupyter, Jira, HTML, CSS

PUBLICATIONS & AWARDS

- **Solaiman, M. S.,** Sakib, M. S. I., Rahman, M. A., & Rashid, A. B. (2024). Effect of cutting parameters on performance of CNC hot wire styrofoam cutting machine. *Proceedings of the 7th Bangladesh International Conference on Industrial Engineering and Operations Management*, Dhaka. [View Publication](#)
- Mahin, A. M. S., **Solaiman, M. S.,** & Shahriar, B. M. (2024). Optimization of inbound logistics of automobile industry using FlexSim. *Proceedings of the 7th Bangladesh International Conference on Industrial Engineering and Operations Management*, Dhaka. [View Publication](#)
- **Award:** 2nd Place, IEOM Undergraduate Student Paper Competition (sponsored by Siemens), 7th IEOM Bangladesh Conference, December 2024, for the first-authored CNC hot wire cutting paper

RESEARCH & PROFESSIONAL EXPERIENCE

Junior Research Assistant Jun 2024 – Dec 2024
Eco Tech Research Lab, Bangladesh

- Investigated how cutting parameters affect the performance of a CNC hot wire styrofoam cutting machine.
- First author of the resulting paper, which won 2nd place in the IEOM Undergraduate Student Paper Competition.

Assistant Engineer, Business Operations & Development Feb 2024 – Dec 2024
Anwar Group of Industries, Bangladesh

- Supported business operations and process-development initiatives across manufacturing units, applying industrial engineering methods to operational problems.

LEADERSHIP & UNIVERSITY SERVICE

Peer Connect Guide Jan 2026 – Present
Adelaide University

- Support and mentor incoming postgraduate students, many of them international, during their transition to university life and study in Adelaide.

President, MIST Career Club Jan 2023 – Jun 2023
Military Institute of Science and Technology (MIST), Dhaka, Bangladesh

- Led club operations, coordinated career-development initiatives and managed collaboration among students, faculty and external stakeholders.